# Untitled

2025-09-12

```
library(BART)
```

```
## Warning: package 'BART' was built under R version 4.4.3
```

```
## Loading required package: nlme
```

```
## Loading required package: survival
```

```
college=read.csv("College.csv",header=T,na.strings="?")
rownames(college) = college[,1]
college=data.frame(college[,-1])
college$Private = as.factor(college$Private)
head(college)
```

```
##                              Private Apps Accept Enroll Top10perc Top25perc
## Abilene Christian University     Yes 1660   1232    721        23        52
## Adelphi University               Yes 2186   1924    512        16        29
## Adrian College                   Yes 1428   1097    336        22        50
## Agnes Scott College              Yes  417    349    137        60        89
## Alaska Pacific University        Yes  193    146     55        16        44
## Albertson College                Yes  587    479    158        38        62
##                              F.Undergrad P.Undergrad Outstate Room.Board Books
## Abilene Christian University        2885         537     7440       3300   450
## Adelphi University                  2683        1227    12280       6450   750
## Adrian College                      1036          99    11250       3750   400
## Agnes Scott College                  510          63    12960       5450   450
## Alaska Pacific University            249         869     7560       4120   800
## Albertson College                    678          41    13500       3335   500
##                              Personal PhD Terminal S.F.Ratio perc.alumni Expend
## Abilene Christian University     2200  70       78      18.1          12   7041
## Adelphi University               1500  29       30      12.2          16  10527
## Adrian College                   1165  53       66      12.9          30   8735
## Agnes Scott College               875  92       97       7.7          37  19016
## Alaska Pacific University        1500  76       72      11.9           2  10922
## Albertson College                 675  67       73       9.4          11   9727
##                              Grad.Rate
## Abilene Christian University        60
## Adelphi University                  56
## Adrian College                      54
## Agnes Scott College                 59
## Alaska Pacific University           15
## Albertson College                   55
```

```
dim(college)
```

```
## [1] 777  18
```

# BART

```
K=10
set.seed(123)
folds=sample(1:K,nrow(college),replace=TRUE)

set.seed(123)
yhat_cv_rmse <- 1:K
for (i in 1:K){
  train <- folds!=i
  x <- college[, -18]
  y <- college[, 18]
  xtrain <- x[train, ]
  ytrain <- y[train]
  xtest <- x[-train, ]
  ytest <- y[-train]

  bartfit <- gbart(xtrain, ytrain, x.test = xtest)
  yhat_cv_rmse[i] <- sqrt(mean((ytest - bartfit$yhat.test.mean)^2))
}
```

```
## *****Calling gbart: type=1
## *****Data:
## data:n,p,np: 709, 18, 776
## y1,yn: -5.242595, 33.757405
## x1,x[n*p]: 0.000000, 4509.000000
## xp1,xp[np*p]: 0.000000, 4509.000000
## *****Number of Trees: 200
## *****Number of Cut Points: 1 ... 100
## *****burn,nd,thin: 100,1000,1
## *****Prior:beta,alpha,tau,nu,lambda,offset: 2,0.95,1.59099,3,30.7158,65.2426
## *****sigma: 12.557291
## *****w (weights): 1.000000 ... 1.000000
## *****Dirichlet:sparse,theta,omega,a,b,rho,augment: 0,0,1,0.5,1,18,0
## *****printevery: 100
##
## MCMC
## done 0 (out of 1100)
## done 100 (out of 1100)
## done 200 (out of 1100)
## done 300 (out of 1100)
## done 400 (out of 1100)
## done 500 (out of 1100)
## done 600 (out of 1100)
## done 700 (out of 1100)
## done 800 (out of 1100)
## done 900 (out of 1100)
## done 1000 (out of 1100)
## time: 9s
## trcnt,tecnt: 1000,1000
## *****Calling gbart: type=1
## *****Data:
## data:n,p,np: 709, 18, 776
## y1,yn: -5.404795, 33.595205
## x1,x[n*p]: 0.000000, 4509.000000
## xp1,xp[np*p]: 0.000000, 4509.000000
## *****Number of Trees: 200
## *****Number of Cut Points: 1 ... 100
## *****burn,nd,thin: 100,1000,1
## *****Prior:beta,alpha,tau,nu,lambda,offset: 2,0.95,1.90919,3,31.7072,65.4048
## *****sigma: 12.758333
## *****w (weights): 1.000000 ... 1.000000
## *****Dirichlet:sparse,theta,omega,a,b,rho,augment: 0,0,1,0.5,1,18,0
## *****printevery: 100
##
## MCMC
## done 0 (out of 1100)
## done 100 (out of 1100)
## done 200 (out of 1100)
## done 300 (out of 1100)
## done 400 (out of 1100)
## done 500 (out of 1100)
## done 600 (out of 1100)
## done 700 (out of 1100)
## done 800 (out of 1100)
## done 900 (out of 1100)
```

```
## done 1000 (out of 1100)
## time: 9s
## trcnt,tecnt: 1000,1000
## *****Calling gbart: type=1
## *****Data:
## data:n,p,np: 700, 18, 776
## y1,yn: -11.580000, 33.420000
## x1,x[n*p]: 0.000000, 4509.000000
## xp1,xp[np*p]: 0.000000, 4509.000000
## *****Number of Trees: 200
## *****Number of Cut Points: 1 ... 100
## *****burn,nd,thin: 100,1000,1
## *****Prior:beta,alpha,tau,nu,lambda,offset: 2,0.95,1.90919,3,31.7889,65.58
## *****sigma: 12.774757
## *****w (weights): 1.000000 ... 1.000000
## *****Dirichlet:sparse,theta,omega,a,b,rho,augment: 0,0,1,0.5,1,18,0
## *****printevery: 100
##
## MCMC
## done 0 (out of 1100)
## done 100 (out of 1100)
## done 200 (out of 1100)
## done 300 (out of 1100)
## done 400 (out of 1100)
## done 500 (out of 1100)
## done 600 (out of 1100)
## done 700 (out of 1100)
## done 800 (out of 1100)
## done 900 (out of 1100)
## done 1000 (out of 1100)
## time: 8s
## trcnt,tecnt: 1000,1000
## *****Calling gbart: type=1
## *****Data:
## data:n,p,np: 706, 18, 776
## y1,yn: -5.291785, 33.708215
## x1,x[n*p]: 0.000000, 4509.000000
## xp1,xp[np*p]: 0.000000, 4509.000000
## *****Number of Trees: 200
## *****Number of Cut Points: 1 ... 100
## *****burn,nd,thin: 100,1000,1
## *****Prior:beta,alpha,tau,nu,lambda,offset: 2,0.95,1.90919,3,32.0056,65.2918
## *****sigma: 12.818222
## *****w (weights): 1.000000 ... 1.000000
## *****Dirichlet:sparse,theta,omega,a,b,rho,augment: 0,0,1,0.5,1,18,0
## *****printevery: 100
##
## MCMC
## done 0 (out of 1100)
## done 100 (out of 1100)
## done 200 (out of 1100)
## done 300 (out of 1100)
## done 400 (out of 1100)
## done 500 (out of 1100)
## done 600 (out of 1100)
## done 700 (out of 1100)
```

```
## done 800 (out of 1100)
## done 900 (out of 1100)
## done 1000 (out of 1100)
## time: 9s
## trcnt,tecnt: 1000,1000
## *****Calling gbart: type=1
## *****Data:
## data:n,p,np: 709, 18, 776
## y1,yn: -5.809591, 33.190409
## x1,x[n*p]: 0.000000, 4509.000000
## xp1,xp[np*p]: 0.000000, 4509.000000
## *****Number of Trees: 200
## *****Number of Cut Points: 1 ... 100
## *****burn,nd,thin: 100,1000,1
## *****Prior:beta,alpha,tau,nu,lambda,offset: 2,0.95,1.90919,3,32.3412,65.8096
## *****sigma: 12.885258
## *****w (weights): 1.000000 ... 1.000000
## *****Dirichlet:sparse,theta,omega,a,b,rho,augment: 0,0,1,0.5,1,18,0
## *****printevery: 100
##
## MCMC
## done 0 (out of 1100)
## done 100 (out of 1100)
## done 200 (out of 1100)
## done 300 (out of 1100)
## done 400 (out of 1100)
## done 500 (out of 1100)
## done 600 (out of 1100)
## done 700 (out of 1100)
## done 800 (out of 1100)
## done 900 (out of 1100)
## done 1000 (out of 1100)
## time: 8s
## trcnt,tecnt: 1000,1000
## *****Calling gbart: type=1
## *****Data:
## data:n,p,np: 705, 18, 776
## y1,yn: -5.590071, 33.409929
## x1,x[n*p]: 0.000000, 4509.000000
## xp1,xp[np*p]: 0.000000, 4509.000000
## *****Number of Trees: 200
## *****Number of Cut Points: 1 ... 100
## *****burn,nd,thin: 100,1000,1
## *****Prior:beta,alpha,tau,nu,lambda,offset: 2,0.95,1.90919,3,32.2657,65.5901
## *****sigma: 12.870199
## *****w (weights): 1.000000 ... 1.000000
## *****Dirichlet:sparse,theta,omega,a,b,rho,augment: 0,0,1,0.5,1,18,0
## *****printevery: 100
##
## MCMC
## done 0 (out of 1100)
## done 100 (out of 1100)
## done 200 (out of 1100)
## done 300 (out of 1100)
## done 400 (out of 1100)
## done 500 (out of 1100)
```

```
## done 600 (out of 1100)
## done 700 (out of 1100)
## done 800 (out of 1100)
## done 900 (out of 1100)
## done 1000 (out of 1100)
## time: 9s
## trcnt,tecnt: 1000,1000
## *****Calling gbart: type=1
## *****Data:
## data:n,p,np: 685, 18, 776
## y1,yn: -5.332847, 33.667153
## x1,x[n*p]: 0.000000, 40386.000000
## xp1,xp[np*p]: 0.000000, 4509.000000
## *****Number of Trees: 200
## *****Number of Cut Points: 1 ... 100
## *****burn,nd,thin: 100,1000,1
## *****Prior:beta,alpha,tau,nu,lambda,offset: 2,0.95,1.8208,3,31.5254,65.3328
## *****sigma: 12.721709
## *****w (weights): 1.000000 ... 1.000000
## *****Dirichlet:sparse,theta,omega,a,b,rho,augment: 0,0,1,0.5,1,18,0
## *****printevery: 100
##
## MCMC
## done 0 (out of 1100)
## done 100 (out of 1100)
## done 200 (out of 1100)
## done 300 (out of 1100)
## done 400 (out of 1100)
## done 500 (out of 1100)
## done 600 (out of 1100)
## done 700 (out of 1100)
## done 800 (out of 1100)
## done 900 (out of 1100)
## done 1000 (out of 1100)
## time: 8s
## trcnt,tecnt: 1000,1000
## *****Calling gbart: type=1
## *****Data:
## data:n,p,np: 689, 18, 776
## y1,yn: -5.307692, 33.692308
## x1,x[n*p]: 0.000000, 4509.000000
## xp1,xp[np*p]: 0.000000, 4509.000000
## *****Number of Trees: 200
## *****Number of Cut Points: 1 ... 100
## *****burn,nd,thin: 100,1000,1
## *****Prior:beta,alpha,tau,nu,lambda,offset: 2,0.95,1.90919,3,31.0171,65.3077
## *****sigma: 12.618731
## *****w (weights): 1.000000 ... 1.000000
## *****Dirichlet:sparse,theta,omega,a,b,rho,augment: 0,0,1,0.5,1,18,0
## *****printevery: 100
##
## MCMC
## done 0 (out of 1100)
## done 100 (out of 1100)
## done 200 (out of 1100)
## done 300 (out of 1100)
```

```
## done 400 (out of 1100)
## done 500 (out of 1100)
## done 600 (out of 1100)
## done 700 (out of 1100)
## done 800 (out of 1100)
## done 900 (out of 1100)
## done 1000 (out of 1100)
## time: 8s
## trcnt,tecnt: 1000,1000
## *****Calling gbart: type=1
## *****Data:
## data:n,p,np: 694, 18, 776
## y1,yn: -5.414986, 33.585014
## x1,x[n*p]: 0.000000, 4509.000000
## xp1,xp[np*p]: 0.000000, 4509.000000
## *****Number of Trees: 200
## *****Number of Cut Points: 1 ... 100
## *****burn,nd,thin: 100,1000,1
## *****Prior:beta,alpha,tau,nu,lambda,offset: 2,0.95,1.90919,3,30.9883,65.415
## *****sigma: 12.612860
## *****w (weights): 1.000000 ... 1.000000
## *****Dirichlet:sparse,theta,omega,a,b,rho,augment: 0,0,1,0.5,1,18,0
## *****printevery: 100
##
## MCMC
## done 0 (out of 1100)
## done 100 (out of 1100)
## done 200 (out of 1100)
## done 300 (out of 1100)
## done 400 (out of 1100)
## done 500 (out of 1100)
## done 600 (out of 1100)
## done 700 (out of 1100)
## done 800 (out of 1100)
## done 900 (out of 1100)
## done 1000 (out of 1100)
## time: 8s
## trcnt,tecnt: 1000,1000
## *****Calling gbart: type=1
## *****Data:
## data:n,p,np: 687, 18, 776
## y1,yn: -5.656477, 33.343523
## x1,x[n*p]: 0.000000, 4509.000000
## xp1,xp[np*p]: 0.000000, 4509.000000
## *****Number of Trees: 200
## *****Number of Cut Points: 1 ... 100
## *****burn,nd,thin: 100,1000,1
## *****Prior:beta,alpha,tau,nu,lambda,offset: 2,0.95,1.90919,3,31.7056,65.6565
## *****sigma: 12.758017
## *****w (weights): 1.000000 ... 1.000000
## *****Dirichlet:sparse,theta,omega,a,b,rho,augment: 0,0,1,0.5,1,18,0
## *****printevery: 100
##
## MCMC
## done 0 (out of 1100)
## done 100 (out of 1100)
```

```
## done 200 (out of 1100)
## done 300 (out of 1100)
## done 400 (out of 1100)
## done 500 (out of 1100)
## done 600 (out of 1100)
## done 700 (out of 1100)
## done 800 (out of 1100)
## done 900 (out of 1100)
## done 1000 (out of 1100)
## time: 9s
## trcnt,tecnt: 1000,1000
```

```
yhat_cv_rmse
```

```
##  [1] 11.17614 10.96139 11.05368 10.96417 11.00650 11.08105 11.02085 11.03371
##  [9] 11.01437 10.96239
```

```
min(yhat_cv_rmse)
```

```
## [1] 10.96139
```

```
mean(yhat_cv_rmse)
```

```
## [1] 11.02742
```

Random Forest ≈ 12.77

XGBosst ≈ 10.8

BART ≈ 11.03